Insurance Platform

API Documentation

Complete REST API Reference v1.0 | March 2026

10	44+	4	JWT
Controllers	Endpoints	User Roles	Auth

Roles: Customer Agent Claim Officer Admin

Overview

The Insurance Platform API provides a comprehensive RESTful interface for managing insurance policies, claims, payments, and user interactions. It supports four user roles, each with different access levels.

Role	Capabilities
Customer	Apply for policies, file claims, make payments, manage notifications
Agent	Review policy applications, manage customers, view analytics & commissions
Claim Officer	Review and approve/reject insurance claims assigned by admin
Admin	Full system access: user management, assignments, dashboard statistics

Authentication

Type: JWT Bearer Token **Header:** `Authorization: Bearer <token>`

Call `POST /api/auth/login` to obtain a token. Include it in the Authorization header for all protected endpoints.

Role	Access	Color
Customer	Policy, Claims, Payments, Notifications, Chat	Green
Agent	Agent ops, assigned policies, analytics, commission	Cyan
ClaimOfficer	Assigned claims, review decisions	Amber
Admin	Full system access, all user & data management	Purple

Controllers Summary

#	Controller	Base Route	Auth	EPs	Purpose
1	AuthController	/api/auth	None	2	User authentication & registration
2	AdminController	/api/admin	Admin	11	System administration & management
3	AgentController	/api/agent	Agent	5	Agent operations & analytics
4	PolicyController	/api/policy	JWT*	4	Policy management & applications
5	ClaimController	/api/claim	Role-based	8	Insurance claim lifecycle
6	ChatController	/api/chat	JWT	4	Real-time customer-agent chat
7	NotificationsController	/api/notifications	JWT	4	User notification management
8	PaymentController	/api/payment	JWT	1	Process policy premium payments
9	ReportController	/api/report	Admin/Agent/Officer	1	Financial & payment reports
10	DashboardController	/api/dashboard	Role-based	4	Dashboard access validation

** /api/policy/configuration is public; all other policy endpoints require JWT.*

1. AuthController

Base Route: /api/auth **Auth:** Not required

Handles user authentication, registration, and JWT token generation.

1.1 Register Customer — POST /api/auth/register

POST **/api/auth/register**

Register a new customer account. Returns JWT token, email, roles, and userId on success.

Field	Type	Required	Description
`fullName`	string	Yes	Customer full name
`email`	string	Yes	Valid email address
`password`	string	Yes	Min 8 chars, upper + lower + digit + symbol
`phoneNumber`	string	Yes	With country code e.g. +919876543210
`aadhaarNumber`	string	No	12-digit Aadhaar number for KYC

Success (200): { status, message, token, email, roles[], userId }

Error (400): { status: Error, message: Email already exists }

1.2 Login — POST /api/auth/login

POST **/api/auth/login**

Authenticate user and receive a JWT bearer token for subsequent requests.

Field	Type	Required	Description
`email`	string	Yes	Registered email address
`password`	string	Yes	User password

Success (200): { status, message, token (JWT), email, roles[], userId }

Error (401): { status: Error, message: Invalid credentials }

2. AdminController

Base Route: /api/admin **Auth:** Admin role required for ALL endpoints

Provides full administrative functions including user management, workload assignment, and system-wide statistics.

2.1 Create Agent — POST /api/admin/create-agent

POST /api/admin/create-agent

Field	Type	Required	Description
`fullName`	string	Yes	Agent full name
`email`	string	Yes	Agent email address
`phoneNumber`	string	Yes	Contact number
`bankAccountNumber`	string	Yes	16-digit account for commission deposits
`password`	string	Yes	Initial login password

2.2 Create Claim Officer — POST /api/admin/create-claim-officer

POST /api/admin/create-claim-officer

Same parameters as Create Agent. Returns userId and email on success.

2.3 – 2.11 Additional Admin Endpoints

Ref	Method	Endpoint	Description
2.3	GET	/api/admin/agents	List all agent accounts
2.4	GET	/api/admin/claim-officers	List all claim officer accounts
2.5	DELETE	/api/admin/delete-user/{id}	Permanently delete a user by ID
2.6	GET	/api/admin/policy-requests	View all policy applications (all statuses)
2.7	GET	/api/admin/agents-with-load	Agent list with assigned policy count
2.8	POST	/api/admin/assign-agent	Assign policy to agent (body: applicationId, agentId)
2.9	GET	/api/admin/all-users	All users: customers, agents, officers
2.10	GET	/api/admin/all-claims	Master view of all insurance claims
2.11	GET	/api/admin/admin-stats	KPI dashboard: users, policies, claims, amounts

Admin Stats Response Fields

Field	Type	Description
totalCustomers	integer	Total registered customers
totalAgents	integer	Total agents in system
totalClaimOfficers	integer	Total claim officers
totalPolicies	integer	All policies (active + pending + others)
activePolicies	integer	Currently active policies
pendingPolicies	integer	Awaiting agent review
totalClaims	integer	All claims filed
pendingClaims	integer	Claims awaiting officer review
approvedClaims	integer	Approved claims count
rejectedClaims	integer	Rejected claims count
totalClaimAmount	decimal	Sum of all requested claim amounts (INR)
approvedClaimAmount	decimal	Sum of all approved claim payouts (INR)

3. AgentController

Base Route: /api/agent **Auth:** Agent role required

Manages agent-specific operations: policy reviews, customer management, commission tracking, and analytics.

Ref	Method	Endpoint	Parameters	Description
3.1	GET	/api/agent/my-requests	None (JWT)	Policy applications assigned to logged-in agent
3.2	POST	/api/agent/review-request	applicationId, status	Approve or reject a policy application
3.3	GET	/api/agent/my-customers	None	List customers with active policies through this agent
3.4	GET	/api/agent/commission-stats	None	Total commission earned + per-policy breakdown
3.5	GET	/api/agent/analytics	None	Full analytics: coverage, portfolio mix, monthly commission chart

3.2 Review Request — Body Parameters

Field	Type	Required	Description
`applicationId`	string	Yes	Policy application ID to review
`status`	string	Yes	"Approved" or "Rejected"

3.5 Analytics Response Fields

Field	Description
totalCoverageProvided	Total sum insured across all active policies (INR)
activePolicyCount	Count of active policies
uniqueCustomerCount	Number of unique customers served
bestPerformingCategory	Category with highest policy count
bestPerformingTier	Tier with highest policy count
totalPremiumCollected	Total premium received (INR)
totalCommissionEarned	Total agent commission earned (INR)
commissionPerformance	Array: { month, value } monthly commission breakdown
portfolioMix	Array: { category, count } by category
tierBreakdown	Array: { tier, count } by tier

4. PolicyController

Base Route: /api/policy **Auth:** JWT required (except /configuration which is public)

Manages insurance policy configuration, premium calculations, applications, and policy viewing.

4.1 Get Policy Configuration — GET /api/policy/configuration (Public)

GET **/api/policy/configuration**

No authentication required. Returns all available policy plans with tiers, pricing, benefits, and risk factor multipliers.

Response Field	Description
policyCategories[]	Available categories: Health Insurance - Individual / Family
categories[].categoryId	Identifier: "individual" or "family"
categories[].maxMembersAllowed	1 for Individual, up to 6 for Family
tiers[].tierId	GOLD or SILVER — used in apply/calculate calls
tiers[].baseCoverageAmount	Base sum insured in INR
tiers[].basePremiumAmount	Annual base premium in INR
tiers[].benefits[]	Array of benefit strings for the tier
riskFactors.ageMultipliers	Age band multipliers: 18-30 = 1.0x, 31-45 = 1.2x, 46-60 = 1.5x
riskFactors.professionMultipliers	Multipliers by profession type
riskFactors.habitAdjustments	Premium adjustments for smoking/drinking habits

4.2 Calculate Premium — POST /api/policy/calculate-premium

POST **/api/policy/calculate-premium**

Returns a calculated premium quote. Does NOT submit an application — use /apply to actually apply.

Field	Type	Required	Description
`policyCategory`	string	Yes	"Health Insurance - Individual" or "Health Insurance - Family"
`tierId`	string	Yes	"GOLD" or "SILVER"
`applicant.fullName`	string	Yes	Applicant's full name
`applicant.age`	integer	Yes	18 to 60 — determines age multiplier
`applicant.profession`	string	Yes	e.g. IT Professional, Farmer, Driver
`applicant.alcoholHabit`	string	Yes	"NonDrinker" | "Occasional" | "Regular"
`applicant.smokingHabit`	string	Yes	"NonSmoker" | "Occasional" | "Regular"

Field	Type	Required	Description
`applicant.travelKmPerMonth`	integer	Yes	Monthly km travelled — affects risk score
`paymentMode`	string	Yes	"Annual" | "Semi-Annual" | "Quarterly" | "Monthly"
`nominee.nomineeName`	string	Yes	Nominee full name
`nominee.nomineeEmail`	string	Yes	Nominee email address
`nominee.nomineePhone`	string	Yes	Nominee phone number
`nominee.nomineeBankAccountNumber`	string	Yes	Nominee bank account for claim payout
`members`	array	No	Family members (required for Family category policy)

4.3 Apply for Policy — POST /api/policy/apply

POST **/api/policy/apply**

Submit a formal policy application. Accepts the same request body as Calculate Premium. Returns applicationId, calculatedPremium, and status: Pending.

4.4 Get My Policies — GET /api/policy/my-policies

GET **/api/policy/my-policies**

Returns all policies for the authenticated customer: Active, Pending, Assigned, Approved, Rejected, Expired.

Field	Type	Description
id	string	Policy application ID
policyCategory	string	Category name
tierId	string	GOLD or SILVER
status	string	Current status of the application
calculatedPremium	decimal	Final calculated premium in INR
totalCoverageAmount	decimal	Total sum insured in INR
startDate	datetime	Policy start date (ISO 8601 UTC)
expiryDate	datetime	Policy expiry date (ISO 8601 UTC)
assignedAgentId	string	ID of assigned agent (null if unassigned)

5. ClaimController

Base Route: /api/claim **Auth:** Role-based per endpoint

Manages the full insurance claim lifecycle: filing, assignment, review, and decision.

Ref	Method	Endpoint	Role	Description
5.1	POST	/api/claim/raise	Customer	File a new claim (multipart/form-data)
5.2	GET	/api/claim/my-claims	Customer	View all claims by the authenticated customer
5.3	GET	/api/claim/admin/pending	Admin	List all pending claims awaiting assignment
5.4	GET	/api/claim/admin/officers	Admin	Officer workload — claim count per officer
5.5	POST	/api/claim/admin/assign	Admin	Assign a claim officer to a pending claim
5.6	GET	/api/claim/officer/my-requests	ClaimOfficer	Get all claims assigned to logged-in officer
5.7	POST	/api/claim/officer/review	ClaimOfficer	Approve or reject a claim with payout amount
5.8	GET	/api/claim/agent/customer-claims	Agent	View claims raised by agent customers

5.1 Raise Claim — Request Fields (multipart/form-data)

Field	Type	Required	Description
`policyApplicationId`	string	Yes	Active policy application ID
`incidentType`	string	Yes	"Accident" | "Sickness" | "Death" | "Property Damage"
`incidentLocation`	string	Yes	Location where incident occurred
`incidentDate`	date	Yes	ISO 8601 date of incident (e.g. 2026-02-25)
`description`	string	Yes	Detailed incident description
`hospitalName`	string	Yes	Name of treating hospital / clinic
`hospitalizationRequired`	boolean	Yes	true if inpatient admission was required
`requestedAmount`	decimal	Yes	Claim amount in INR
`affectedMemberName`	string	No	Family policy: name of affected member
`affectedMemberRelation`	string	No	"Spouse" | "Son" | "Daughter" etc.
`documents`	File[]	No	PDF/JPG/PNG files — max 10 files, 5 MB each

5.7 Review Claim — Request Body

Field	Type	Required	Description
`claimId`	string	Yes	Claim ID to review
`status`	string	Yes	"Approved" or "Rejected"
`approvedAmount`	decimal	Yes	Amount approved for payout (INR); 0 if Rejected
`remarks`	string	No	Officer notes or justification

6. ChatController

Base Route: /api/chat **Auth:** JWT required

Enables real-time messaging between customers and their assigned agents around specific policy applications.

Ref	Method	Endpoint	Body / Params	Description
6.1	GET	/api/chat/list	None	Get all chat conversations for the authenticated user
6.2	GET	/api/chat/{policyId}	policyId (URL)	Retrieve full message history for a specific policy chat
6.3	POST	/api/chat/init	policyId (body)	Initialize a new chat session for a policy application
6.4	POST	/api/chat/{policyId}/read	policyId (URL)	Mark all messages in a policy chat as read

7. NotificationsController

Base Route: /api/notifications **Auth:** JWT required

Manages user notifications triggered by policy status changes, claim updates, and system events.

Ref	Method	Endpoint	Description
7.1	GET	/api/notifications	Get all notifications for the authenticated user
7.2	GET	/api/notifications/unread-count	Returns integer count of unread notifications
7.3	POST	/api/notifications/{id}/read	Mark a specific notification as read by its ID
7.4	POST	/api/notifications/read-all	Mark all notifications as read in one call

8. PaymentController

Base Route: /api/payment **Auth:** JWT required

8.1 Process Payment — POST /api/payment/process

POST **/api/payment/process**

Process the premium payment for an approved policy application. All amounts in INR.

Field	Type	Required	Description
`applicationId`	string	Yes	The approved policy application ID to pay for
`amount`	decimal	Yes	Payment amount in INR — must match calculatedPremium

Success (200): { status: Success, message: Payment processed, paymentId, policyStatus: Active }

9. ReportController

Base Route: /api/report **Auth:** Admin / Agent / ClaimOfficer

9.1 Unified Payment Report — GET /api/report/unified-payments

GET **/api/report/unified-payments**

Role-filtered financial report. No query parameters required.

Role	Data Returned
Admin	All payments and claim payouts across the entire platform
Agent	Payments related to the agent's customer portfolio
ClaimOfficer	Claim payout transactions handled by this officer

10. DashboardController

Base Route: /api/dashboard **Auth:** Role-based

Returns 200 OK if the caller has the correct role; 403 Forbidden otherwise. Used for frontend access validation.

Ref	Method	Endpoint	Required Role
10.1	GET	/api/dashboard/customer	Customer
10.2	GET	/api/dashboard/agent	Agent
10.3	GET	/api/dashboard/claim-officer	ClaimOfficer
10.4	GET	/api/dashboard/admin	Admin

Error Handling

All errors use a standard JSON format: `{ status, message, errors[] }`

HTTP Code	Scenario	Example Message
400 Bad Request	Validation failed	"Age must be between 18 and 60"
400 Bad Request	Business rule violation	"Email already exists"
401 Unauthorized	Missing or expired JWT	"Invalid or expired token"
403 Forbidden	Wrong role for endpoint	"You do not have permission to access this resource"
404 Not Found	Resource does not exist	"Resource not found"
429 Too Many Requests	Rate limit exceeded (100/min/user)	"Too many requests, please slow down"
500 Internal Server Error	Unhandled server error	"An unexpected error occurred"

HTTP Status Codes

Code	Status	Description
200	OK	Request successful
201	Created	Resource created successfully
400	Bad Request	Invalid request parameters or validation error
401	Unauthorized	Missing or invalid authentication token
403	Forbidden	Authenticated but not authorized for this resource
404	Not Found	Resource not found
429	Too Many Requests	Rate limit exceeded
500	Internal Server Error	Server error occurred

Additional Notes

Topic	Details
Base URL	https://api.insuranceplatform.com
Rate Limiting	100 requests per minute per user/IP. Returns 429 when exceeded.
File Uploads	Supported: PDF, JPG, JPEG, PNG. Max 5 MB per file, max 10 files per claim. Use multipart/form-data on POST /api/claim/raise.

Topic	Details
Date / Time	All datetime fields use ISO 8601 format (e.g. 2026-03-03T08:30:00Z). Timezone: UTC.
Currency	All monetary values in INR (Indian Rupees). Decimal with 2 places e.g. 18000.00.
Content-Type	application/json for all endpoints except file upload endpoints (multipart/form-data).
CORS	Configure allowed origins on the server for browser-based clients.

Complete Endpoint Reference — All 44 Endpoints

Every endpoint across all 10 controllers. Use with base URL: https://api.insuranceplatform.com

#	Method	Endpoint	Auth	Description
1	POST	/api/auth/register	None	Register new customer, returns JWT
2	POST	/api/auth/login	None	User login, returns JWT
3	POST	/api/admin/create-agent	Admin	Create a new agent account
4	POST	/api/admin/create-claim-officer	Admin	Create a new claim officer
5	GET	/api/admin/agents	Admin	List all agents
6	GET	/api/admin/claim-officers	Admin	List all claim officers
7	DELETE	/api/admin/delete-user/{id}	Admin	Delete a user by ID
8	GET	/api/admin/policy-requests	Admin	View all policy applications
9	GET	/api/admin/agents-with-load	Admin	Agent workload overview
10	POST	/api/admin/assign-agent	Admin	Assign agent to policy
11	GET	/api/admin/all-users	Admin	List all users
12	GET	/api/admin/all-claims	Admin	View all claims (master)
13	GET	/api/admin/admin-stats	Admin	Dashboard KPI statistics
14	GET	/api/agent/my-requests	Agent	Get assigned policy requests
15	POST	/api/agent/review-request	Agent	Approve or reject a policy
16	GET	/api/agent/my-customers	Agent	List agent customers
17	GET	/api/agent/commission-stats	Agent	Commission earned
18	GET	/api/agent/analytics	Agent	Performance analytics
19	GET	/api/policy/configuration	None (Public)	Policy plans, tiers, risk factors
20	POST	/api/policy/calculate-premium	JWT	Calculate premium quote
21	POST	/api/policy/apply	JWT	Submit policy application

#	Method	Endpoint	Auth	Description
22	GET	/api/policy/my-policies	JWT	View my policies
23	POST	/api/claim/raise	Customer	File a new insurance claim
24	GET	/api/claim/my-claims	Customer	View my claims
25	GET	/api/claim/admin/pending	Admin	List pending claims
26	GET	/api/claim/admin/officers	Admin	Officer workload
27	POST	/api/claim/admin/assign	Admin	Assign officer to claim
28	GET	/api/claim/officer/my-requests	ClaimOfficer	Get assigned claims
29	POST	/api/claim/officer/review	ClaimOfficer	Approve or reject claim
30	GET	/api/claim/agent/customer-claims	Agent	View customer claims
31	GET	/api/chat/list	JWT	Get all chat conversations
32	GET	/api/chat/{policyId}	JWT	Get policy chat history
33	POST	/api/chat/init	JWT	Initialize chat session
34	POST	/api/chat/{policyId}/read	JWT	Mark chat messages as read
35	GET	/api/notifications	JWT	Get all notifications
36	GET	/api/notifications/unread-count	JWT	Get unread notification count
37	POST	/api/notifications/{id}/read	JWT	Mark one notification as read
38	POST	/api/notifications/read-all	JWT	Mark all notifications as read
39	POST	/api/payment/process	JWT	Process policy premium payment
40	GET	/api/report/unified-payments	Admin/Agent/Officer	Unified payment report
41	GET	/api/dashboard/customer	Customer	Customer dashboard access

#	Method	Endpoint	Auth	Description
42	**GET**	`/api/dashboard/agent`	Agent	Agent dashboard access
43	**GET**	`/api/dashboard/claim-officer`	ClaimOfficer	Claim officer dashboard
44	**GET**	`/api/dashboard/admin`	Admin	Admin dashboard access